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The bending model of the Transdanubian Central Range (Hungary) in the light of Triassic palaeomagnetic data

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SUMMARY

In the Carpatho–Pannonian region, the Transdanubian Central Range is the second largest unit with changing strike along the SW–NE-trending Mesozoic sequence. This structure was tested palaeomagnetically for oroclinal bending with the help of Triassic data, most of them derived from widespread platform carbonates. Based on nearly 700 samples, it was possible to obtain a statistically well-defined overall mean palaeomagnetic direction for a Werfenian–early Carnian age group in the western part of the Range, and a different direction for a late Carnian–Rhaetian age group, extending over the whole length of the Range. These two mean directions are $D=316^\circ$, $I=38^\circ$ ($k=70$, $\alpha_{95}=8^\circ$, $n=9$) and $D=292^\circ$, $I=57^\circ$ ($k=31$, $\alpha_{95}=10^\circ$, $n=10$), respectively. The statistical parameters of these overall mean palaeomagnetic directions are comparable to those characterizing rigid areas. Nevertheless, an attempt was made to correlate the tectonic trend and palaeomagnetic declination, which lead to rejecting the model of oroclinal bending. However, it was possible to separate the late Carnian–Rhaetian sampling localities into a western (west of Lábatlan) and an eastern group, each of them with a statistically good overall mean direction: $D=312^\circ$, $I=56^\circ$ ($k=139$, $\alpha_{95}=6^\circ$, $n=5$) for the western, and $D=273^\circ$, $I=54^\circ$ ($k=51$, $\alpha_{95}=11^\circ$, $n=5$) for the eastern segment. The good agreement of the mean inclinations, together with an approximately 40° difference in mean declinations indicates that the Transdanubian Central Range east of Lábatlan has undergone a greater counterclockwise rotation than the western part; i.e. there is a 'break' in the Transdanubian Central Range between Lábatlan and Dorog (at the Tertiary Dorog Basin). Thus, the relative rotation between the western and eastern segments is in the opposite sense to that required by oroclinal bending.

Key words: oroclinal bending, palaeomagnetism.

INTRODUCTION

In orogenic belts, the tectonic strike often changes in a systematic manner, giving the impression of an arc. Some structures of this kind call for models that account for the strike trends observed on a regional scale by secondary bending of an originally elongated, straight sedimentary trough (oroclinal bending; Carey 1958). Secondary bending involves vertical-axis rotations that change in sense and angle in a systematic way and match the variation of the strike along the bend.

Palaeomagnetism offers a unique method for testing arcuate structures for oroclinal bending. However, the theoretically straightforward test is, in practice, difficult for several reasons. First, the number of sites/samples collected from penecontemporaneous rocks must be quite high, with the sites/samples distributed so that they provide a good control on the structure. Furthermore, local structural control at the sampling sites is essential; this means not only well-defined and measurable stratification, but quantitative information on tectonic plunge. Even if such information is available, it is best to avoid sampling steeply inclined strata, because of the danger of declination error (Márton 1977; MacDonald 1980), which remains less than 5° (10°) when the dip angle of the bed is less than 30° (40°), no matter how steep the plunge may be (Tarling 1983). Needless to say, slump structures are to be avoided. The next problem is the isolation of a single-component, statistically well-defined mean palaeomagnetic direction for each site in the test. Finally, the age of the magnetization must be constrained. An acceptable practice is to demonstrate the consistency of palaeomagnetic inclinations, since a full fold test at each sampling site would be an impossible condition to satisfy.

The test performed on a single 'age-window' of pre-orogenic rocks may be decisive, provided that the window is narrow in relation to the speed of the apparent polar wander for the tectonic unit in question. The significance, however, can be enhanced by testing two or more pre-orogenic age-windows.

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In the Carpatho–Pannonian region, apart from the well-known Carpathian arc, the Transdanubian Central Range is the largest unit with changing strike along the SW–NE-trending Mesozoic sequence (Vígh & Szentes 1952): the strike is SW–NE in the Keszthelyi Hills, the Bakony Mts and the Vértes Hills, is practically E–W in the Gerecse Hills, and turns to NW–SE in the Pilis and Budai Hills as well as east of the Danube river, in the area of a Transdanubian Range-type Mesozoic sequence (Fig. 1).

Balla (1988), following Wein (1976), was of the opinion that the 'kink' in the strike is due to oroclinal bending. He attempted to prove this by making use of the Jurassic–early Cretaceous palaeomagnetic results of Márton & Márton (1983, 1985), data that were never intended to test the Transdanubian Range for bending. To the contrary, Márton & Márton (1983) demonstrated that the apparent polar wander was fast during the late Jurassic–early Cretaceous for the Transdanubian Central Range as well as for various other tectonic units of southern Tethyan origin; these authors also concluded that significant relative movements could not have taken place within and between the Bakony, Gerecse and Vértes areas.

In the data set available to Balla (1988) there were no observations east of the 'kink', and the number of penecontemporaneous data west of the 'kink' was insufficient to constrain the model. Therefore, Balla recalculated early Jurassic palaeomagnetic directions to match the late Jurassic–early Cretaceous ones, using average values of declination and inclination differences (due to apparent polar wander) between the two age groups. Unfortunately, in the time window he selected, the speed of the declination shift was high (Márton & Márton 1983); that is, differences in declinations could have been caused by age differences.

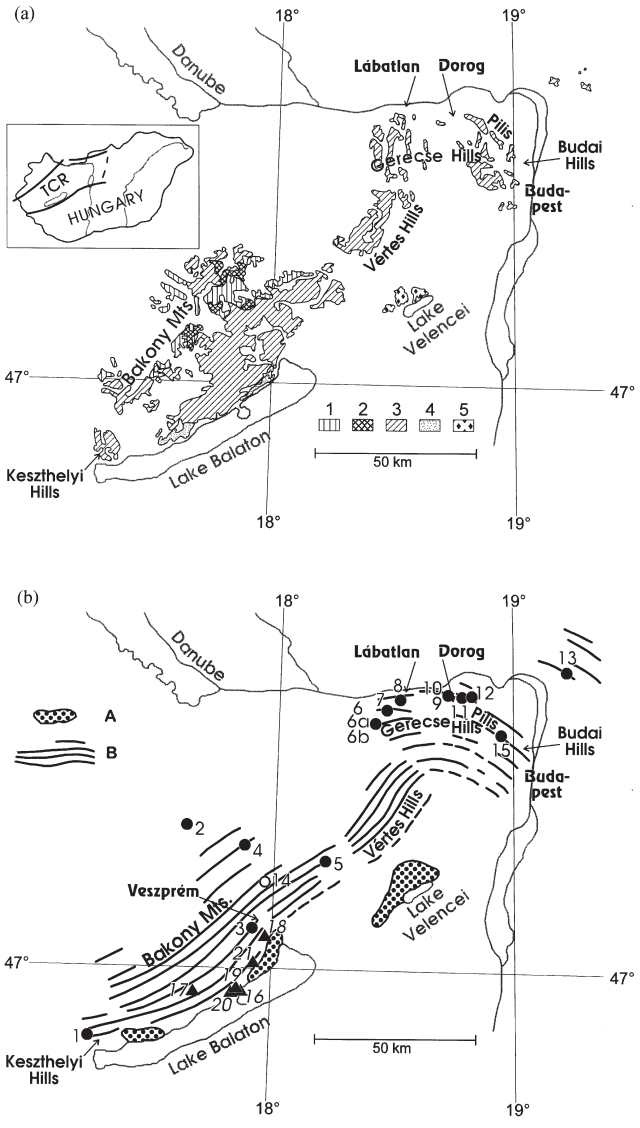


Figure 1. (a) Geological sketch map of the Transdanubian Central Range and the Transdanubian-type Triassic east of the River Danube. (The location of the study area in Hungary is shown in the upper left corner.) Key to geology: 1: Cretaceous, 2: Jurassic, 3: Triassic, 4 Palaeozoic sediments, 5: Palaeozoic granite. (b) Triassic sampling localities from the Transdanubian Central Range-type Mesozoic. Base map modified after Balla (1988) (A: Palaeozoic, B: oroclinal trends in the Mesozoic). Triangles: Werfenian to early Carnian; dots: late Carnian to Rhaetian localities. Numbers refer to Tables 1 and 2 and are used throughout the text.

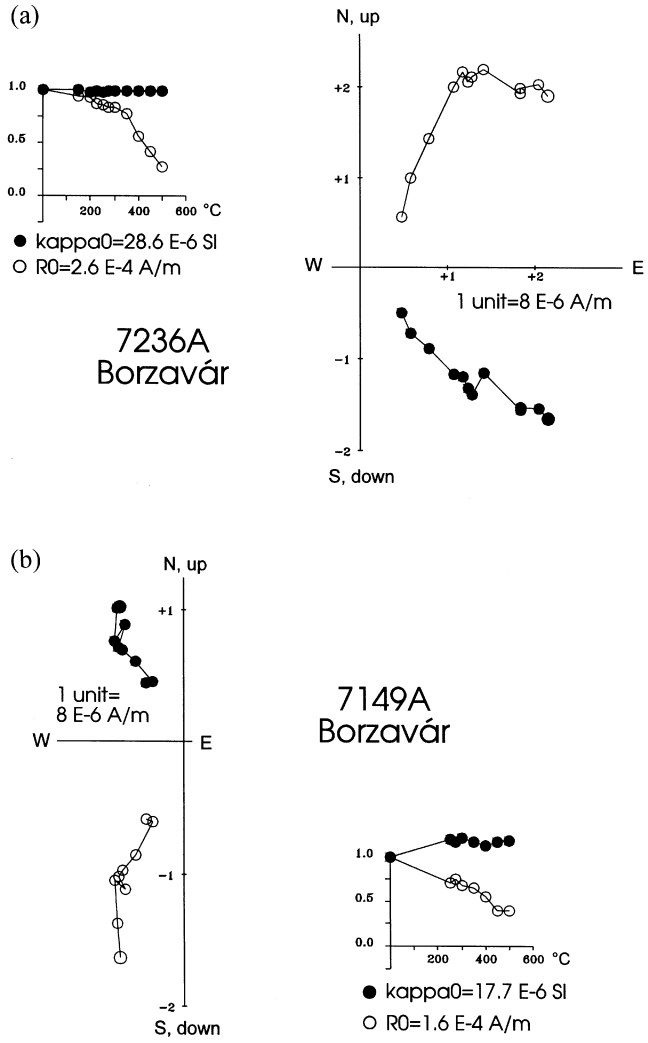


Figure 2. West of Lábatlan (locality 4): examples of stepwise thermal demagnetization. Orthogonal plots and relative intensity/susceptibility versus temperature curves. Samples with reversed (2a: 7236A) and normal (2b: 7149A) polarity (for more examples, see Márton & Haas 1996).

The easiest way to prove or disprove the validity of the bending model for the Transdanubian Range would have been to add Jurassic–early Cretaceous data to the existing ones, east of the 'kink'. Unfortunately, there is only a single rather poor outcrop of suitable age in that part of the structure. Therefore, to test the bending model properly, an older age group had to be investigated, namely the Triassic, since the late Triassic carbonates cropout along the whole length of the structure.

This paper presents new palaeomagnetic results from the Triassic that are relevant to testing the bending model.

GEOLOGY AND SAMPLING

In the Transdanubian Central Range, the most widespread rock type is the thick (2–2.5 km) platform carbonate sequence of Carnian through Rhaetian age. The sequence was deposited on a quiet shelf, after equilibration (basins filled with terrigenous material) of the Mid-Triassic dissected topography. The late Carnian through middle Norian of this region is diagenetically dolomitized (Main Dolomite); the younger part is practically free of dolomitization (Dachstein Limestone). Lofer cycles (Fischer 1964) characterize the whole sequence, with 'B' (intertidal) and 'C' (subtidal) members in the Main Dolomite, and complete cycles, including palaeosol horizons ('A' member) in the Dachstein Limestone (Haas 1994, 1995). The presence of palaeosol horizons is very important since it increases the chance of preserving the primary remanent magnetization (Márton & Haas 1996).

In general, the Main Dolomite is followed by Dachstein Limestone, in which dolomitization is subordinate and is restricted to the 'B' member of the Lofer cycle. The difference in dolomite content is explained by climatic change from arid (during the formation of the Main Dolomite) to humid (during deposition of the Dachstein Limestone; Haas 1995). The Main Dolomite and Dachstein Limestone also occur as heteropic facies during the Norian; that is, the Main Dolomite was still forming in the Bakony Mts when the Dachstein Limestone commenced deposition in the NE part of the Transdanubian Range. Further to the east, east of the Danube (Csővár area), Carnian through Rhaetian pelagic limestone was deposited synchronously with the formation of platform carbonates (Haas 1988). In the southwestern part of the Transdanubian Range, Kössen facies (anoxic, argillaceous limestone, deposited

Table 1. Triassic platform carbonates and heteropic facies, west and east of the River Danube. Palaeomagnetic results. Directions to be interpreted in terms of tectonics are shown in heavy print.

	locality	n	D°	I°	k	α_{95}°	D_C°	I_C°	polarity	dip
1.	Rezi *Norian (m-u)*	8	**291**	**49**	89	6	218	70	N	142/42
2.	Ugod, Szárhegy *Norian (u)*	11	**292**	**45**	16	12	258	55	N	159/28
3.	Veszprém, Aranyosvölgy *Carnian (u)*	16	**276**	**58**	43	6	288	35	N	304/22
4.	Borzavár, quarry *Rhaetian (l)*	28	324	50	46	4	**318**	**55**	mix	180/7
5.	Bodajk	6	**314**	**50**	11	21	319	14	R	328/37
6.	T5 deep drill,Tata *Rhaetian*	33	290	54	30	5	**300**	**60**	mix	76/10
6.a	Tata, cavity infillings *Rhaetian*	30 (7)	312	58	194	4	**325**	**62**	N	76/10
7.	Agostyán, Gorbabánya *Rhaetian*	30	300	52	26	5	**307**	**52**	mix	32/5
8.	Lábatlan, Kecskekő quarry *Rhaetian*	58	306	47	43	3	**313**	**52**	mix	80/8
9.	Dorog, quarry Nagykőszikla 1. *Rhaetian*	12	248	55	41	7	**265**	**50**	N	327/15
10.	Dorog, quarry Nagykőszikla 2. *Rhaetian*	9	239	53	198	4	**253**	**58**	N	6/11
11.	Kesztölc quarry *Norian*	17	274	44	29	7	**290**	**49**	N	30/16
12.	Pilisszentlélek, Feketehegy *Norian (m-l)*	11	236	56	26	9	**261**	**57**	N	varied tilts max. dip angle 32
13.	Csővár *Rhaetian*	12	281	47	20	10	**291**	**49**	N	26/9
15.	Kiskevély *Norian*	10	**287**	**38**	11	15	304	30	N	15/30

Key. n: number of samples on which the statistics are based, (except 6a, where the statistics are based on the number of cavity infillings from which the samples were drilled); D, I, D_C, I_C: declination, inclination before and after tilt correction; k and α_{95}: statistical parameters (Fisher 1953). Mean palaeomagnetic directions accepted for tectonic interpretation are in heavy print. Polarity: N – normal; R – reversed; Mix – mixed. Working quarries are indicated by the word 'quarry' in the respective entries.

Table 2. Werfenian–early Carnian palaeomagnetic results from the Transdanubian Central Range.

	locality	n	D°	I°	k	α_{95}°	D_c°	I_c°	polarity	dip
16.	Balatonhenye, quarry, *Anisian*	12	246	44	32	8	**304**	**38**	N	0/58
17.	Felsőörs, Malom valley, *Anisian*	33	306	62	13	7	**304**	**41**	mix*	298/22
18.	Pécsely, quarry *Carnian*	32	315	36	27	5	**320**	**39**	N	75/7
19.	Balatonfüred Száka-hill, quarry *Werfenian*	9	320	34	55	7	**315**	**38**	N	208/8
20.	Csopak, Nosztory valley, *Carnian*	9	328	66	228	3	**314**	**33**	mix*	310/36
21.	Vászoly *Carnian*	6	342	41	278	4	**336**	**38**	N	267/7

*only one sample is reversed

Key: see Table 1.

Table 3. Reversal test (McFadden & McElhinny 1990).

Locality		Normal polarity					Reversed polarity						
		n	D°	I°	k	α_{95}°	n	D°	I°	k	α_{95}°	ang. separation	critical angle
4.	Borzavár	8	315	+50	42	9	13	146	-50	67	5	7.6	9.6
6.	Tata, T5	27	289	+53	28	6	6	120	-52	27	13	6.4	13.2
7.	Agostyán	16	310	+53	62	4	6	107	-51	78	6	10.8	7.0
8.	Lábatlan	35	306	+48	62	3	10	112	-47	65	6	10.0	7.0

Key: see Table 1.

Table 4. Fold test (McFadden 1990).

	"unfolding" %	D°	I°	k	α_{95}°	ξ_1
late Carnian - Rhaetian localities: 4-8 critical ξ 95%: 2.609	0	306	53	84.7	8.4	2.271
	100	312	56	139.4	6.5	1.452
	min. ξ: 100					
	max. k: 112	313	57	140.5	6.5	
late Carnian - Rhaetian localities: 9-13 critical ξ 95%: 2.609	0	257	52	33.8	13.3	2.136
	100	273	54	50.7	10.8	0.803
	min. ξ: 100					
	max. k: 131	278	53	52.2	10.7	
Werfenian - early Carnian localities: 16-21 critical ξ 99%: 3.919	0	311	51	10.0	22.3	4.377
	100	315	38	70.5	8.0	0.629
	min. ξ: 98	315	39	69.6	8.1	0.041
	max. k: 105	316	37	71.5	8.0	

Key. D: declination; I: inclination; k and α_{95}: statistical parameters.

in a restricted basin) characterize the upper Norian–lower Rhaetian (Haas 1994), in contrast to the open platform depositional environment of the Main Dolomite and the Dachstein Limestone.

According to Haas & Császár (1987), intrashelf rifting took place in the Jurassic and the area subsided. The breaking up of the platform was accompanied by tilting and differential subsidence between blocks. The Cretaceous was the time of alternating periods of epirogenic subsidence and uplift with 'remarkable tectonic movements' in the early Albian and the

Table 5. Overall palaeomagnetic mean directions for different age windows of the Mesozoic for the Transdanubian Central Range.

		Transdanubian Central Range		
		W of Lábatlan	all	E of Lábatlan
1	Werfenian - early Carnian	316/38 k=70 α_{95}=8 N=9		
2	late Carnian - Rhaetian „primary remanence"	312/56 k=139 α_{95}=6 N=5	292/57 k=31 α_{95}=10 N=10	273/54 k=51 α_{95}=11 N=5
3	late Carnian - Rhaetian „secondary remanence"	294/51 k=54 α_{95}=13 N=4	292/49 k=49 α_{95}=11 N=5	287/38* k=11 α_{95}=15 n=10
4	early Jurassic	305/53 k=68 α_{95}=8 N=6		
5	late Jurassic „primary remanence"	268/38 k=100 α_{95}=7 N=5		
6	late Jurassic „secondary remanence"			257/43* k=21 α_{95}=11 n=10
7	Aptian	284/46 k=132 α_{95}=11 N=3		
8	Albian	300/38 k=71 α_{95}=15 N=3		

* single locality

Key. D/I: declination/inclination; k and α_{95}: statistical parameters; N: number of localities; n: number of samples.
Lines 1–3: calculated from Tables 1 and 2 of this paper; Line 5: Márton (1986); Lines 7–8 Márton & Márton (1983).

early Senonian. During the first period, the major part of the Transdanubian Range was elevated, eroded and gently folded, and the general synclinal shape of the Range developed. Regional emergence and denudation marked the second phase (early Senonian). Based on 'regional considerations', Wein (1976) and Balla (1988) suggested that the arc of the Range was formed before the Tertiary, probably during the mid-Cretaceous.

The samples analysed for this study were collected from Werfenian through Rhaetian sediments. Some of the localities represent restricted basins where anoxic conditions must have prevailed (localities 1, 12, 16, 17). Main Dolomite was collected at two points (localities 3 and 5); localities 2, 4, 6, 6a, 7, 8, 9, 10, 11, 14, 15 are in open platform facies (locality 2 is characterized by an abundance of broken-up parts of the previously deposited 'A' and 'B' horizons; and at localities 14 and 15 'A' horizons are not observable); finally, localities 13, 18, 19, 20, 21 are in pelagic limestones and marls.

The samples were drilled and magnetically oriented in the field. In large quarries, several tens of samples were taken, especially when the lithology was varied; that is, when all members of the carbonate platform were well-developed, and dolomitization and reworking of the sediments were subordinate (Fig. 1, Tables 1 and 2: localities 4, 6a, 7, 8, 9, 10, 11, 18), as such platform carbonates are the most likely to preserve depositional–early diagenetic magnetization. Sampling locality 6 is a special case, because the samples were taken from a 200 m drill core, stored in proper order and correct upright position. The latter was checked by J. Haas, who inspected each of the palaeomagnetically tested pieces of the thick core for sedimentological way-up features. The thick core was opened perpendicular to the strike, and the palaeomagnetic samples were oriented first relative to the dip direction. Finally, the position of the magnetic vector was restored with the help of the tectonic dip measured in the outcrop at the top of the drill hole (locality 6a). Despite the subhorizontal dip (angle 10°), this method for re-orienting the drill core samples was highly successful (see statistical parameters for locality 6 in Table 1).

Localities 2, 3, 5, 14 and 15 were sampled to investigate platform carbonates, which are clearly not the best candidates for preserving primary remanence but may carry a magnetic signal that is still useful for providing constraints on the tectonic model of the present study.

MEASUREMENTS AND RESULTS

Normally, a pair of sister specimens of standard size were cut from each core. All specimens were subjected to NRM and susceptibility measurements using a two-component cryogenic magnetometer, JR-4 spinners with a very low noise level and a KLY-2 Kappabridge.

All samples were weakly magnetic. Extremely low NRM intensity and diamagnetic susceptibility characterized the 'C' members of the platform carbonates (for more detailed information on the magnetic properties of Triassic platform carbonates with Lofer cycles see Márton & Haas 1996).

At least one specimen per sample was thermally demagnetized in increments of 20–50 °C starting at 150 °C. To monitor possible changes in magnetic mineralogy, susceptibility was remeasured after each heating step. (The sister specimen was often AF demagnetized, but this procedure very rarely yielded a good result.) After the removal of an overprint component, the NRM direction decayed towards the origin (Figs 2–4). These lined trajectories served as the basis for calculating the mean directions shown in Tables 1 and 2).

The magnetic minerals in the Triassic open platform carbonates of the Transdanubian Range are magnetite with or without goethite; in 'A' and 'B' members, haematite may also occur (for more details, see Márton & Haas 1996). In the other rock types, the carrier of the characteristic magnetization seems to be magnetite, although at some localities the IRM acquisition

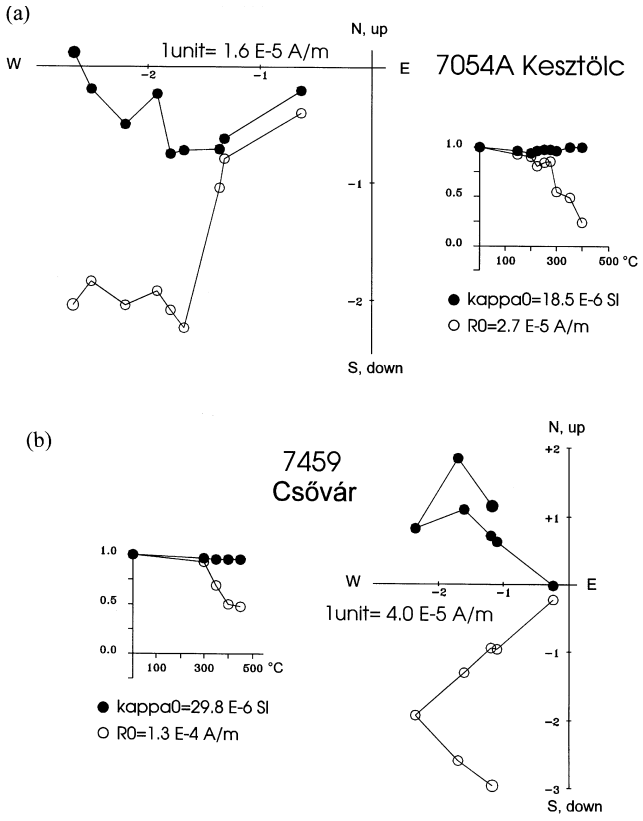


Figure 3. East of Lábatlan: examples of stepwise thermal demagnetization. Orthogonal plots and relative intensity/susceptibility versus temperature curves. A sample from a platform-type limestone (3a: 7054A, locality 11) and another from pelagic limestone (3b: 7459, locality 13). The last demagnetization step of the latter sample is an AF step of 40 mT.

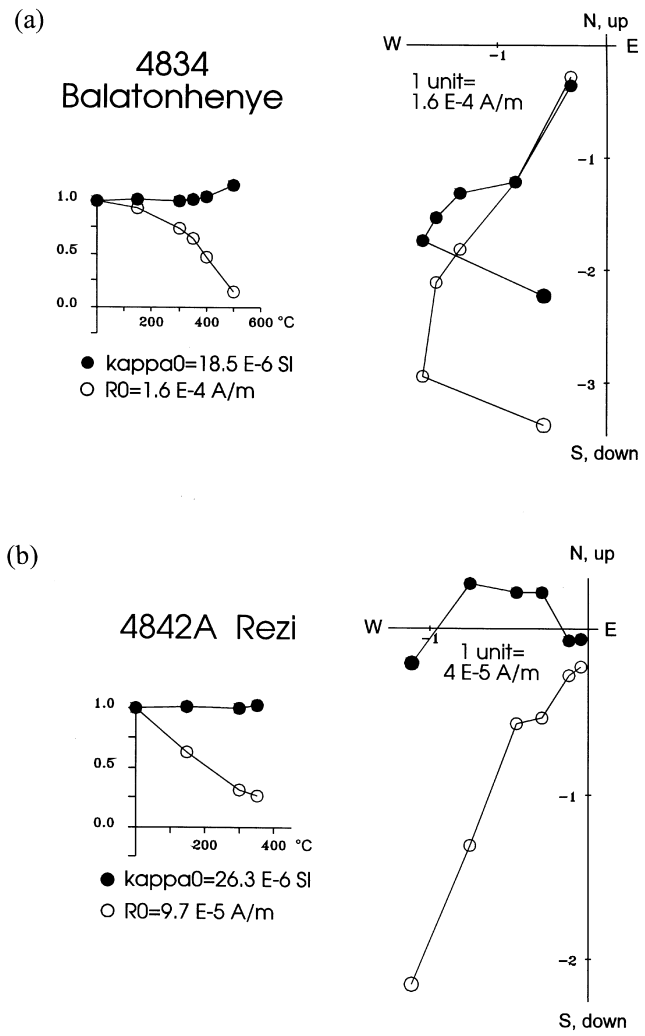


Figure 4. West of Lábatlan: two localities representing restricted basin carbonates. Examples of stepwise thermal demagnetization. Orthogonal plots and relative intensity/susceptibility versus temperature curves. Sample 4834 (4a) from an Anisian locality (16), sample 4842A (4b) from a Norian locality (1).

diagrams and the decay characteristics of the three-component IRM on thermal demagnetization (Fig. 5) reveal the presence of goethite (e.g. sample 4830B) or minor haematite (7463B) in addition to the dominant magnetite. It is worth mentioning the lack of any evidence for iron sulphides, even in samples representing the restricted basins (e.g. specimens 4830B and 4842B in Fig. 5).

At some localities, all samples yielded characteristic remanence. At others, the calculation of the mean direction is based on less than the collected number of samples. In platform carbonates, typically 20–30 per cent of the samples were too weakly magnetic to yield well-defined demagnetization curves. About one-third of these were readily measurable in the natural state, but the intensity of the NRM dropped to the noise level of the magnetometers at 150 °C, and the results therefore were rejected.) With one exception (locality 14), the remaining samples were sufficient in number to allow the calculation of a locality mean direction, and only two localities (5 and 15) exhibit poor statistical parameters due to extremely weak remanence. The results from locality 6b were excluded, because they exhibited remagnetization circle distributions.

The number of rejected samples is lower for the older Triassic.

DISCUSSION AND CONCLUSIONS

The most remarkable feature of the data set presented in Tables 1 and 2 is that the palaeomagnetic mean directions depart significantly from the local direction of the present geomagnetic field ($D=0°$, $I=63°$), both before and after tilt correction. This fact indicates that the NRMs are ancient and were acquired before significant tectonic rotations ended.

All the locality mean directions, both before and after tilt correction, exhibit counterclockwise declination rotation. The inclinations for the late Carnian–Rhaetian carbonates are consistent (a practical criterion for penecontemporaneous magnetizations) in the tectonic system, except for localities 1, 3, 5 and 15 (Table 1). These localities, together with locality 2, are also suspected of carrying late diagenetic or secondary (but still ancient!) remanence for a variety of reasons. Locality 1 is a bituminous limestone; thus, magnetite may well be younger than the stratigraphic age; locality 2 contains a large number of broken-up pieces of previously deposited platform sediment, and the fragments of the 'A' members, reworked during storms are magnetized consistently with the non-reworked material; localities 3 and 5 are thoroughly dolomitized; and at locality 15, 'A' members are missing. Therefore, the characteristic

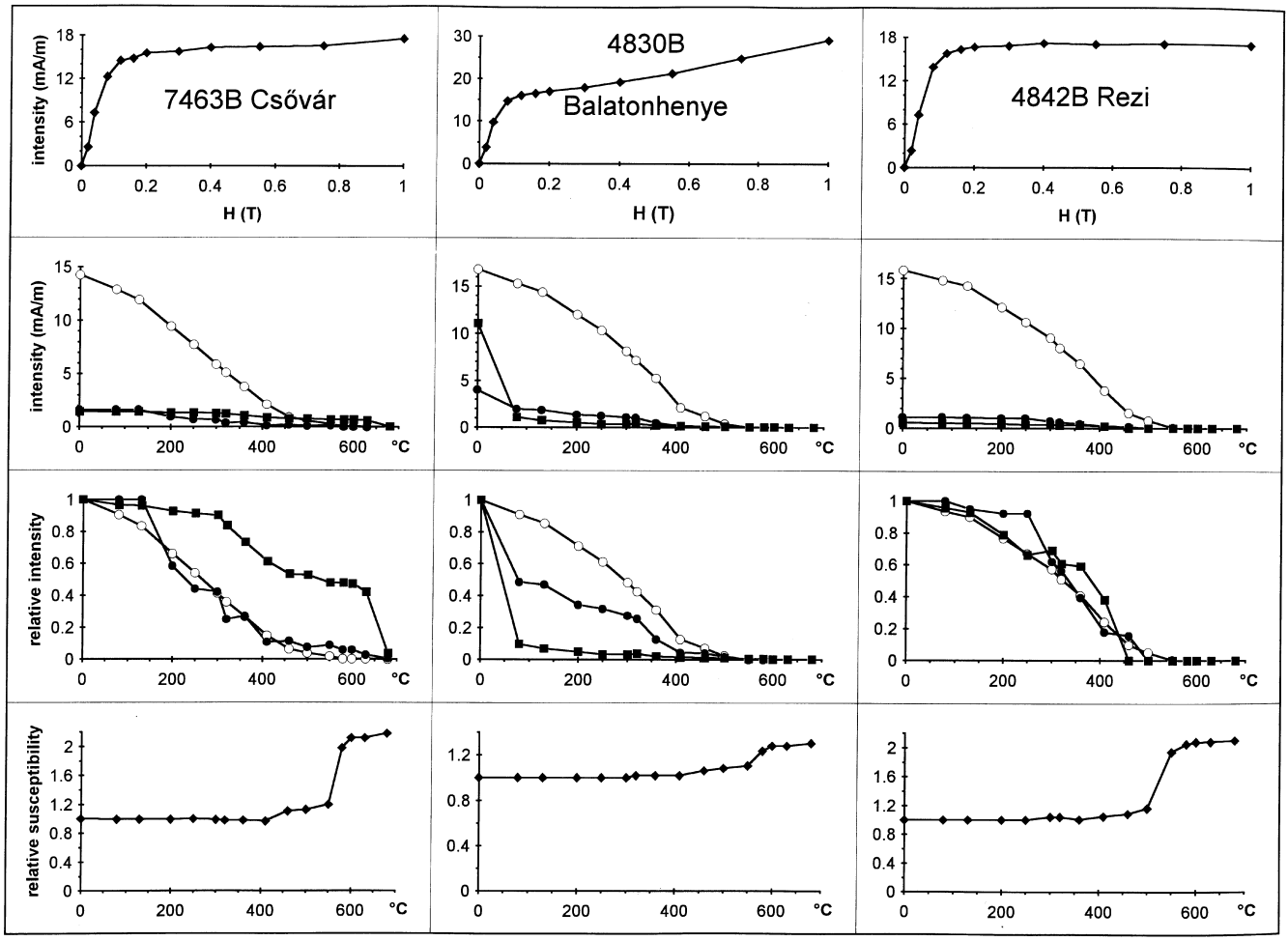


Figure 5. Magnetic identification of the potential carriers of the NRM. From top to bottom: IRM acquisition curves; intensity of the three-component IRM (soft: open circles; medium: solid circles; hard: squares) on thermal demagnetization (method of Lowrie 1990); same normalized to the initial values; normalized susceptibility measured after each heating step. Hard, medium and soft components were acquired in fields of 1.0, 0.36 and 0.2 T, respectively. Since original susceptibilities were diamagnetic, 80×10^{-6} SI was added to the measured susceptibilities before normalization.

magnetizations of these localities are interpreted to be of post-tilting age.

Concerning the remaining localities from the late Carnian–Rhaetian sediments, there are observations supporting the 'primary' age of their characteristic remanences. Although a classical fold test could not be carried out because of a lack of outcrop-scale folding, other evidence points to a depositional–early diagenetic age for the characteristic remanence. This evidence includes: (1) the fairly tight clustering of directions within a locality, despite differences in lithology and magnetic mineralogy for different members of the platform carbonates; (2) antipodal directions, when both polarities are present (Table 3), or good correlation of magnetic parameters and polarity patterns between subsections at the same locality; and (3) the random orientation of the characteristic remanence directions in rip-up breccia horizons (for locality 8, see Márton & Haas 1996).

For the Werfenian–early Carnian group of localities, the depositional–early diagenetic age of the characteristic magnetizations is supported by a positive tilt test on a regional scale (Table 4). The locality mean inclinations characterizing this group are significantly shallower than those of the younger Triassic localities with 'primary' remanence (Fig. 6). Fig. 6 also reveals that, among the younger Triassic localities, some exhibit similar, and some different declinations from those of the older Triassic localities.

The younger age window includes locality mean directions from sites along the whole length of the 'arc', and the older one supports the western segment. When correlation is attempted between tectonic trend and palaeomagnetic declination, the failure is quite obvious (Fig. 7). However, the declinations do show a definite spatial distribution: namely that the declinations for localities 4–8 are rotated less in the counterclockwise sense than those for localities 9–13 (Table 4). It is to be noted that for the first set as well as for the second, restoring the beds to the horizontal improves the statistical parameters (Table 4).

Between the western and eastern segments, the closeness of the overall-mean inclinations is remarkable. Therefore, the 40° difference in declinations calls for tectonic interpretation. Since declination is even more westerly in the eastern segment than in the western part, we can safely conclude that palaeomagnetic data contradict the bending model advocated by Balla (1988).

The 'break' of the declinations between Lábatlan and Dorog,

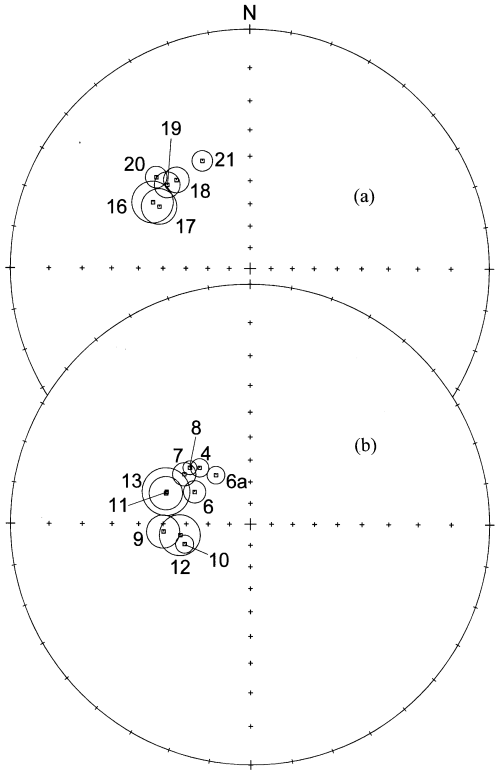


Figure 6. 'Primary' Triassic palaeomagnetic locality mean directions with α_{95}. Stereographic projection, all directions are in the lower hemisphere. (a) Werfenian through early Carnian, all west of Lábatlan. (b) Late Carnian through Rhaetian. 4–8 west of Lábatlan. 9–13 East of Lábatlan.

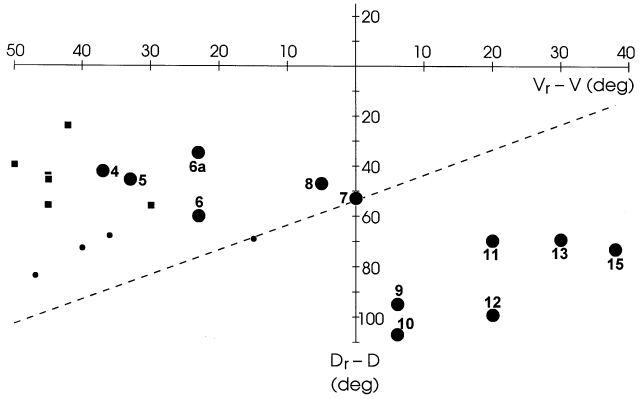


Figure 7. Transdanubian Central Range. Plot of declination deviation Dr − D and the vergency deviation Vr − V for the late Carnian–Rhaetian localities with pre-tilting (large dots), and post-tilting (small dots) magnetizations, and for Werfenian–early Carnian localities with pre-tilting remanences. Reference vergency is V = 0°, reference declination is D = 0°. Dashed line is the correlation line between vergency and declination.

with larger counterclockwise rotation in the east, is not easily fitted into existing tectonic models, which emphasize the primary importance of NE–SW-trending tectonic features in the Pannonian Basin. Palaeomagnetic observations, however, exist for the Tertiary that point to a profound difference in the rotation patterns of the Bakony–Vértes–Gerecse area and the North Hungarian Palaeogene Basin, with no clear boundary between them (Márton & Márton 1995; Márton *et al.* 1996). It would be premature to identify the 'break' in the Triassic declination pattern with a Tertiary tectonic boundary, but this possibility cannot be excluded either. The Triassic 'palaeomagnetic' boundary may be aligned with a tectonic line shown by Brezsnyánszky (1989) to be an important feature and also with one of the seismologically most active lines of the Transdanubian Central Range today (Szabó & Páncsics, 1997).

In the regional context of the Alps–Pannonian–Carpathian system, the new palaeomagnetic results are also significant, since they undermine a strong argument for the juxtaposition of the Transdanubian Central Range and the Northern Calcareous Alps throughout the Mesozoic, as was suggested by Balla (1988). Late Jurassic–early Cretaceous palaeomagnetic results (Mauritsch & Márton 1995), as well as the results of the present study, suggest that a direct tectonic connection between the Transdanubian Range and the Northern Calcareous Alps after the Triassic is unlikely.

Regarding the results of the present study in the general tectonic frame of the large plates bordering the Mediterranean, the Triassic inclinations reflect the shift of the continents to the north during the Triassic. Compared to younger (but still 'pre-bending') palaeomagnetic data from the Transdanubian Central Range, the late Triassic inclinations are as steep as the early Jurassic ones, but become shallower in the late Jurassic–Neocomian (Table 5) because of the opening of the Tethys Ocean, with the Transdanubian Central Range at the southern rim.

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REFERENCES

Balla, Z., 1988. Clockwise palaeomagnetic rotations in the Alps in the light of the structural pattern of the Transdanubian Range (Hungary), *Tectonophysics,* **145,** 277–292.

Brezsnyánszky, K., 1989. *Orogenic Structures of Hungary and its Environs,* Geological Institute of Hungary.

Carey, S.W., 1958. A tectonic approach to continental drift, in *Continental Drift Symposium,* pp. 177–355, ed. Carey, S.W., University of Hobart, Tasmania.

Fisher, R.A., 1953. Dispersion on a sphere, *Proc. R. Soc. Lond.,* A, **217,** 127–149.

Fischer, A.G., 1964. The Lofer cyclothems of the Alpine Triassic. Kansas, *Geol. Surv. Bull.,* **169,** 102–149.

Haas, J., 1988. Upper Triassic carbonate platform evolution in the Transdanubian Mid-Mountains, *Acta Geol. Hung.,* **31** (3–4), 299–312.

Haas, J., 1994. Lofer cycles of the Upper Triassic Dachstein platform in the Transdanubian Mid-Mountains, Hungary, *Spec. Publ. Int. Ass. Sediment.,* **19,** 303–322.

Haas, J., 1995. Upper Triassic platform carbonates in the Northern Bakony Mts, *Földtani Közlöny,* **125** (1–2), 27–64.

Haas, J. & Császár, G., 1987. Cretaceous of Hungary: paleogeographic implication, *Rend. Soc. Geol. It.,* **9,** 203–208.

Lowrie, W., 1990. Identification of ferromagnetic minerals in a rock

by coercivity and unblocking temperature properties. *Geophys. Res. Lett.,* **17,** 159–162.

MacDonald, W.D., 1980. Net rotation, apparent tectonic rotation and the structural tilt correction in paleomagnetic studies, *J. geophys. Res.,* **85,** 3659–3669.

McFadden, P.L., 1990. A new fold test for palaeomagnetic studies, *Geophys. J. Int.,* **103,** 163–169.

McFadden, P.L. & McElhinny, M.W., 1990. Classification of the reversal test in palaeomagnetism, *Geophys. J. Int.,* **103,** 725–729.

Márton, E., 1986. The problems of correlation between magnetozones and Calpionellid zones in late Jurassic–early Cretaceous sections, *Acta Geol. Hung.,* **29,** (1–2), 125–131.

Márton, P., 1977. Basic principles of the tectonic application of palaeomagnetism, *Magyar Geofizika,* **XVIII,** 161–165 (in Hungarian).

Márton, E. & Haas, J., 1996. Ancient platform carbonates with well-developed Lofer cyclicity: new candidates for magnetostratigraphy and geodynamically oriented palaeomagnetism, *Geophys. J. Int.,* **126,** 253–262.

Márton, E. & Márton, P., 1983. A refined polar wander curve for the Transdanubian Central Mountains and its bearing on the Mediterranean tectonic history, *Tectonophysics,* **98,** 43–57.

Márton, E. & Márton, P., 1985. Tectonic and palaeomagnetic aspects of palaeomagnetism studies in the Transdanubian Central Mountains, *Acta Geol. Hung.,* **28,** (1–2), 59–70.

Márton, E. & Márton, P., 1995. Large scale rotations in North Hungary during the Neogene as indicated by palaeomagnetic data, in *Palaeomagnetism and Tectonics of the Mediterranean Region,* eds Morris A. & Tarling, D.H., Geol. Soc. Spec. Publ. No. 105, 153–173.

Márton, E., Vass, D. & Túnyi, I., 1996. Rotation of the North Hungarian Paleogene and Lower Miocene rocks indicated by paleomagnetic data (S. Slovakia, N–NE. Hungary), *Geologica Carpathica,* **47,** 31–41.

Mauritsch, H. & Márton, E., 1995. Escape models of the Alpine–Carpathian–Pannonian region in the light of palaeomagnetic observations, *Terra Nova,* **7,** 44–50.

Szabó, Z. & Páncsics, Z., 1997. Gravity lineaments and earthquake distribution in Hungary, *Magyar Geofizika,* **38,** 16–24 (in Hungarian).

Tarling, D.J., 1983. *Paleomagnetism. Principles and Applications in Geology, Geophysics and Archeology,* Chapman and Hall, London.

Vígh, F. & Szentes, F., 1952. Structural and protection-layer conditions in the Dorog coal basin with a special reference to the prevention of the cavern water afflux, *Bányász. Lapok,* **85** (11), 588–600 (in Hungarian).

Wein, Gy., 1976. Die Entwicklungsgeschichte der Budaer Gebirges, *Acta Geol. Acad. Sci. Hung.,* **20** (1–2), 135–160.